Synoptic CB: Macrophyte Fluxes

2022 Samples

2026-03-16

Contents

#Read in and collate all the files in the Processed Data folder

```
ch4 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CH4.csv")
ch4$X <- NULL
ch4 <- ch4 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("ch4_", .)
  ) %>%
  rename(ch4_qaqc_flag = qaqc_flag)

co2 <- read.csv("Processed Data/COMPASS_Synoptic_CB_Macrophyte_Fluxes_Final_2022_CO2.csv")
co2$X <- NULL
co2 <- co2 %>%
  rename_with(
    .cols = 7:(ncol(.) - 4),
    .fn = ~paste0("co2_", .)
  ) %>%
  rename(co2_qaqc_flag = qaqc_flag)

# Combine all files
all_dat <- merge(ch4, co2, by = c("Plot", "Site", "Zone", "Replicate", "Date", "TIMESTAMP", "TIMESTAMP_
```

##Pull in Hydrago probe data

```
## Some flux IDs are missing from hydrago data

##  [1] "GCW_WC_1_202206" "GCW_WC_1_202207" "GCW_WC_1_202208" "GCW_WC_1_202209"
##  [5] "GCW_WC_2_202206" "GCW_WC_2_202207" "GCW_WC_2_202208" "GCW_WC_2_202209"
##  [9] "GCW_WC_3_202206" "GCW_WC_3_202207" "GCW_WC_3_202208" "GCW_WC_4_202206"
```

```
## [13] "GCW_WC_4_202207" "GCW_WC_4_202208" "GCW_WC_5_202206" "GCW_WC_5_202207"
## [17] "GCW_WC_5_202208" "GWI_TR_1_202207" "GWI_TR_1_202209" "GWI_TR_1_202211"
## [21] "GWI_TR_2_202206" "GWI_TR_2_202207" "GWI_TR_2_202209" "GWI_TR_2_202211"
## [25] "GWI_TR_3_202206" "GWI_TR_3_202207" "GWI_TR_3_202209" "GWI_TR_3_202211"
## [29] "GWI_TR_4_202206" "GWI_TR_4_202207" "GWI_TR_4_202209" "GWI_TR_4_202211"
## [33] "GWI_TR_5_202206" "GWI_TR_5_202207" "GWI_TR_5_202209" "GWI_TR_5_202211"
## [37] "GWI_WC_1_202206" "GWI_WC_1_202207" "GWI_WC_2_202206" "GWI_WC_2_202207"
## [41] "GWI_WC_2_202209" "GWI_WC_3_202206" "GWI_WC_3_202207" "GWI_WC_3_202209"
## [45] "GWI_WC_4_202206" "GWI_WC_4_202207" "GWI_WC_4_202209" "GWI_WC_5_202206"
## [49] "GWI_WC_5_202207" "GWI_WC_5_202209" "MSM_WC_1_202209" "MSM_WC_1_202210"
## [53] "MSM_WC_1_202211" "MSM_WC_2_202206" "MSM_WC_2_202209" "MSM_WC_2_202210"
## [57] "MSM_WC_2_202211" "MSM_WC_3_202206" "MSM_WC_3_202209" "MSM_WC_3_202210"
## [61] "MSM_WC_3_202211" "MSM_WC_4_202206" "MSM_WC_4_202209" "MSM_WC_4_202210"
## [65] "MSM_WC_4_202211" "MSM_WC_5_202206" "MSM_WC_5_202209" "MSM_WC_5_202210"
## [69] "MSM_WC_5_202211"
```

##Remove bad fluxes and calculate how many there were

```
## Number flux measurements removed: 2
```

```
## Number negative CO2 fluxes removed: 0
```

```
## Number messy fluxes removed: 2
```

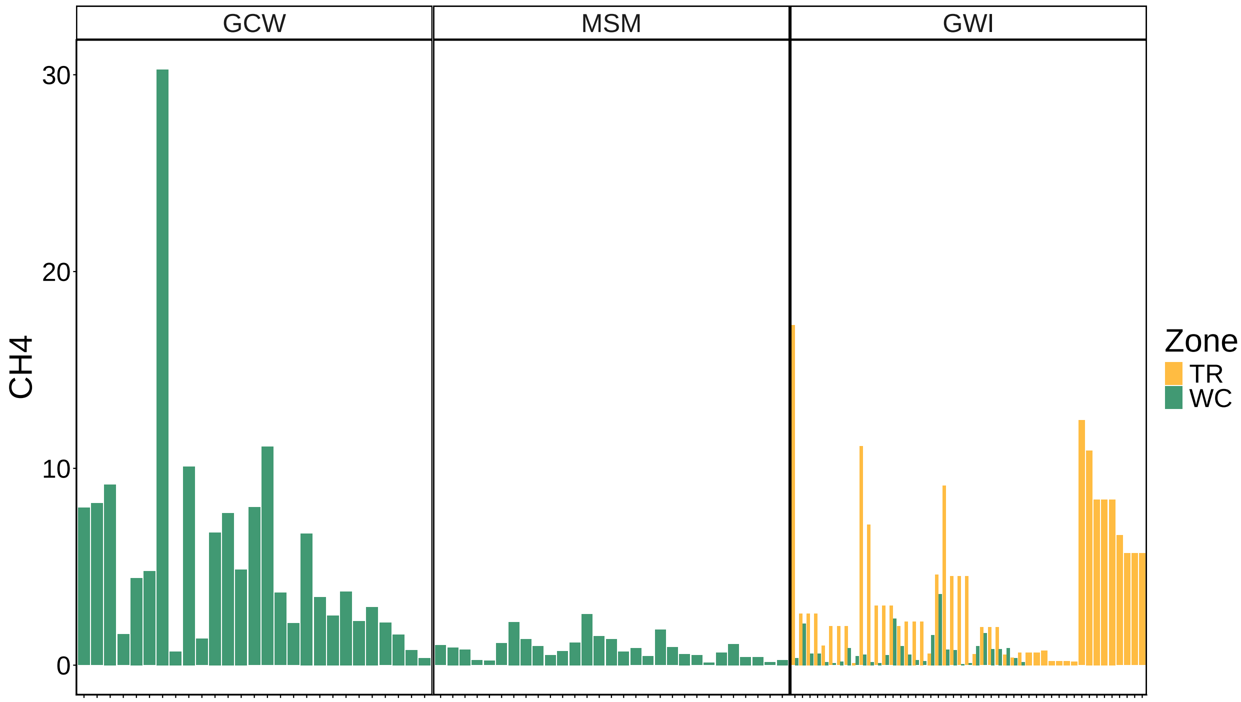
```
## Percentage flux measurements removed: 1.74%
```

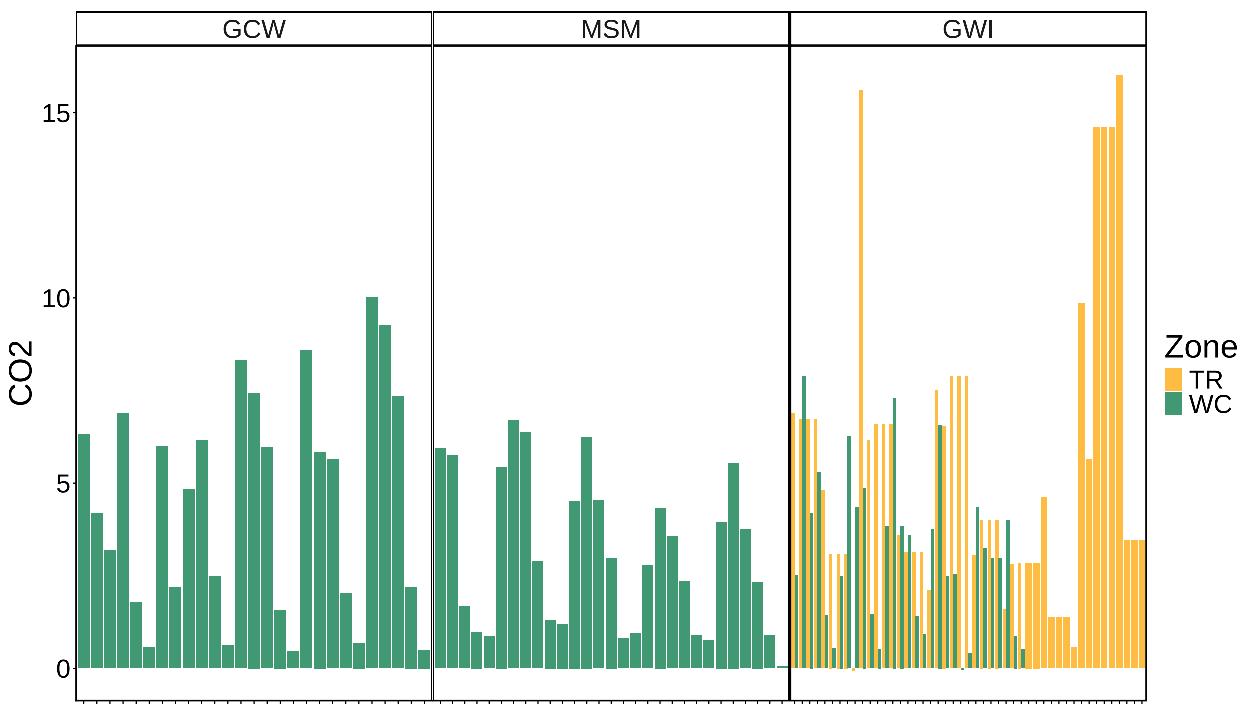
```
## Percentage negative CO2 fluxes removed: 0%
```

```
## Percentage messy fluxes removed: 1.74%
```

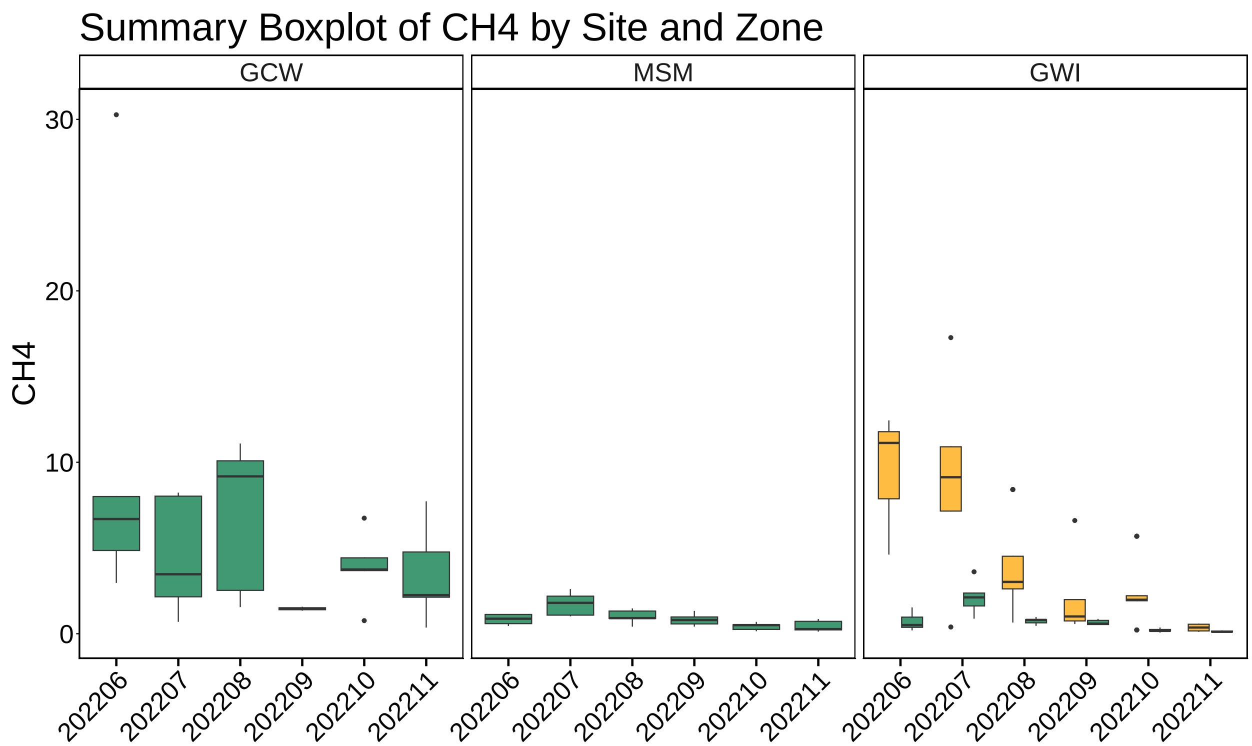
0.1 Write out files

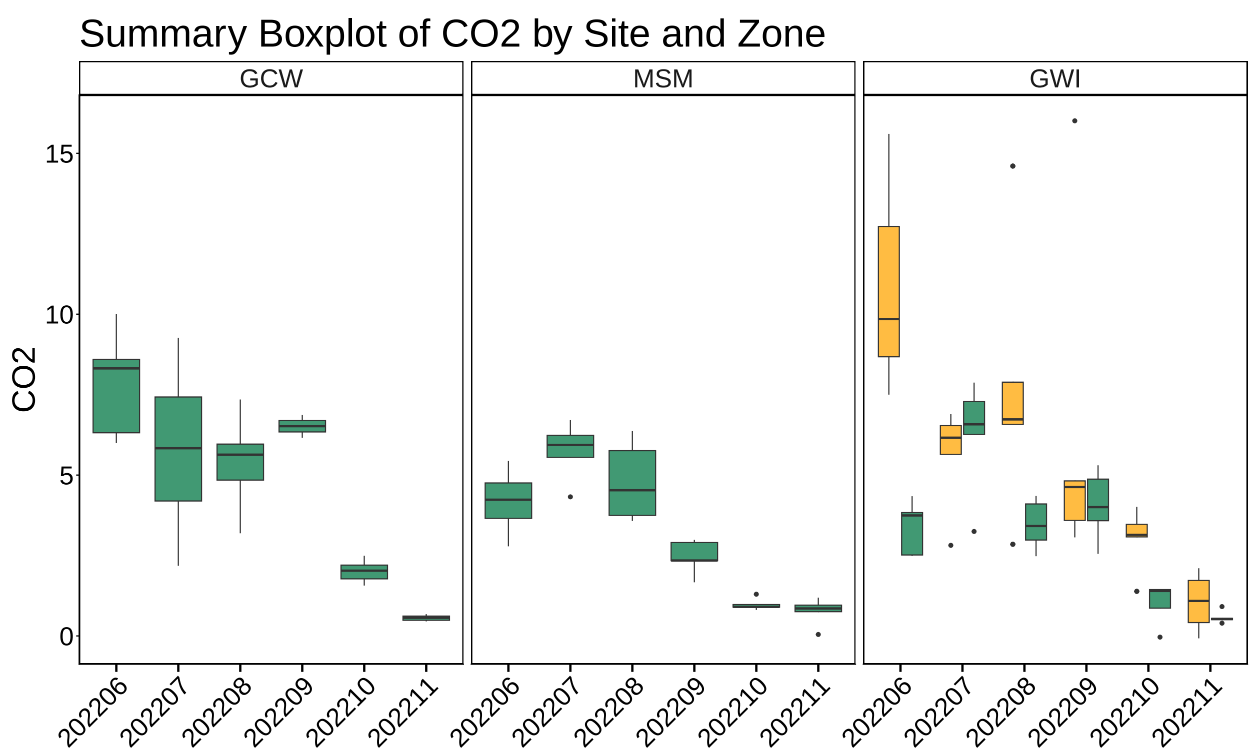
0.2 All Data by Year, Site and Zone



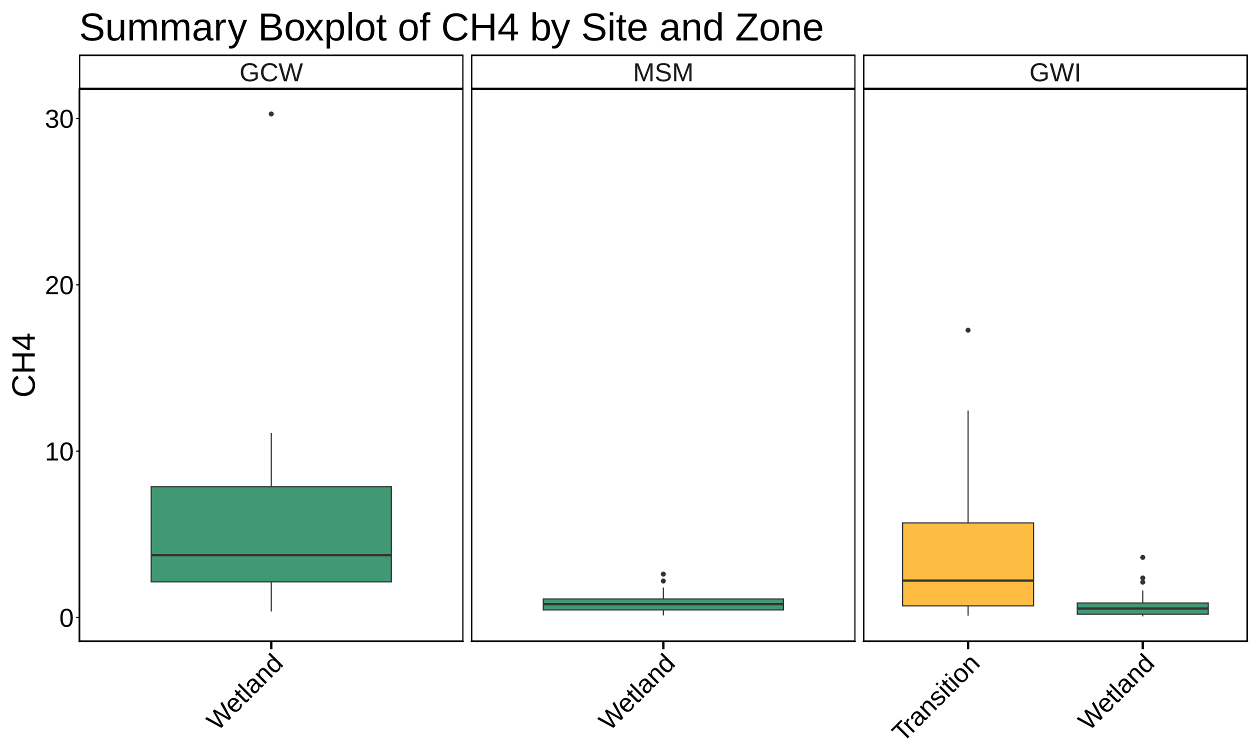


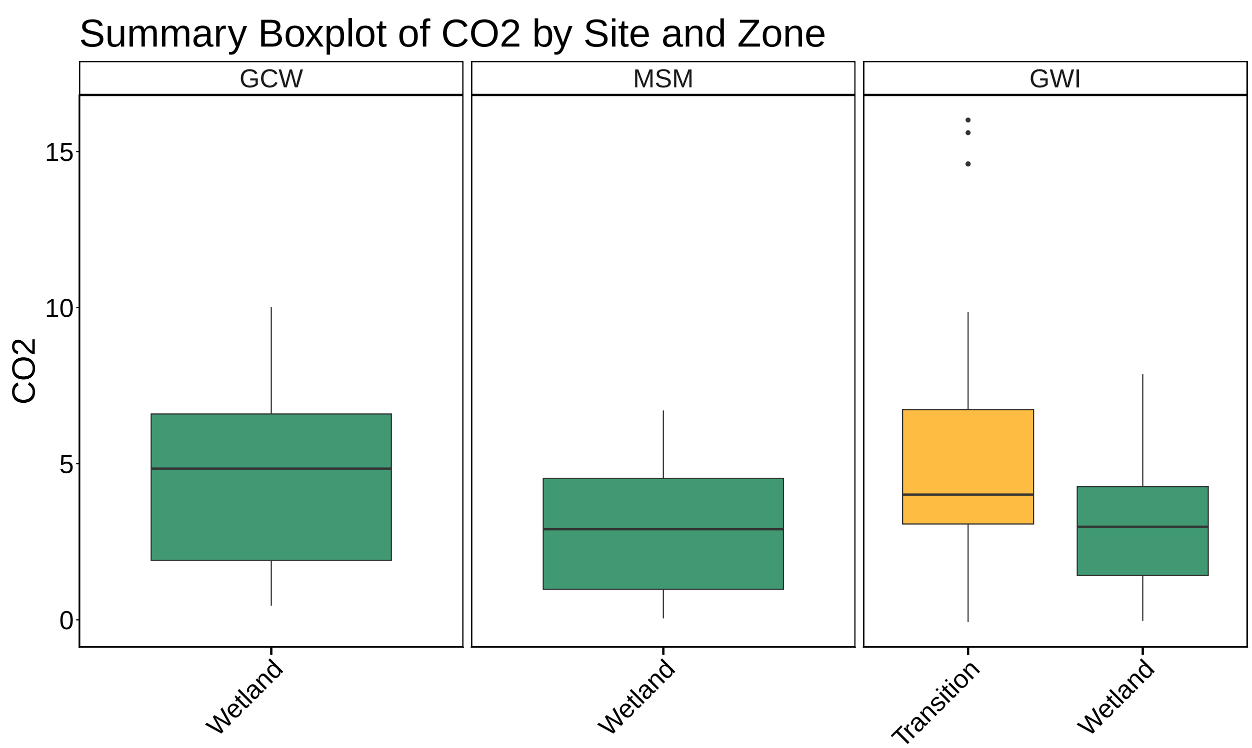
0.3 Boxplot summary by Date





0.4 Boxplot summary by Site





##Calculate summary metrics

0.5 Summary Tables

Table 1: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
GCW	Wetland	0.3585206	30.270376	5.5296620	0.4518196	10.011470	4.481874
MSM	Wetland	0.1222875	2.606256	0.8783515	0.0458841	6.706775	3.115307
GWI	Transition	0.1071858	17.271153	3.8105971	-0.0745288	16.009112	5.500315
GWI	Wetland	0.0653984	3.612942	0.7609969	-0.0364413	7.873652	3.139441